



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	DR . Jayaram Boga
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Roll Number	2520030509
Branch	CSE
Course Outcome	CO-3

Case Study Topic: Zombie Infection Spread Simulation using Breadth First Search (BFS)

Work Done Summary:

In this case study, we implemented **Breadth First Search (BFS)** for simulating zombie infection spread in a survival game network.

The algorithm was used to predict how infection spreads from one infected zone to nearby zones within a limited number of stages. We inserted safe zones and infection paths into a graph structure and applied BFS traversal to identify all affected zones efficiently. The infection network contains overlapping paths where the same zone may be infected through multiple routes. To avoid duplicate infection counting, we implemented a **visited-set mechanism** using Java HashSet.

Implementation Details:

1. Created graph vertices representing survival zones.
2. Added directed edges representing infection spread paths.
3. Implemented Breadth First Search (BFS) traversal from infected source zone A.
4. Limited traversal depth to 3 stages of infection spread.
5. Used Queue data structure for BFS processing.
6. Used HashSet as visited-set to prevent duplicate infection counting.
7. Verified overlapping infection path handling during traversal.
8. Concluded BFS efficiently simulates infection spread prediction in survival systems

```
Zone: A | Infection Depth: 0
Zone: B | Infection Depth: 1
Zone: C | Infection Depth: 1
Zone: D | Infection Depth: 2
Zone: E | Infection Depth: 2
Zone: F | Infection Depth: 2
Zone: G | Infection Depth: 3
Zone: H | Infection Depth: 3
Zone: I | Infection Depth: 3
```

Total Infected Zones: 9

- BFS Traversal:
 - $O(V + E)$
Where:
 - V = Number of zones
 - E = Number of infection paths
- Compared to brute-force traversal methods, BFS provides efficient infection spread analysis.

GitHub Link: <https://github.com/venkatalaxmisamhita/DSA-II-PROJ>

Student Signature	Faculty Signature